Influence, Brokerage, and Community Recovery in the Political Blogosphere: A Centrality and Community-Detection Analysis of the Political Blogs Network

***Abstract*—This report investigates how influence and brokerage manifest in the hyperlink network of political blogs surrounding the 2004 U.S. presidential election, and evaluates whether the network's link structure alone recovers the two known political orientations (liberal and conservative). Using the Political Blogs dataset of Adamic and Glance, we construct a cleaned directed graph of 1,222 blogs and 19,021 hyperlinks, compute two taught centrality measures (PageRank and Betweenness Centrality), design a centrality-informed node-removal experiment contrasting targeted hub removal against random removal, apply Louvain modularity maximization for community detection, and evaluate the detected communities against ground-truth political labels using a pairwise confusion-matrix methodology. We find that link structure alone recovers political orientation with high accuracy (0.884) and F1 score (0.878) on the unmodified network. Critically, removing only the top 5% of blogs by PageRank authority degrades this recovery substantially (F1 falls to 0.684) and fragments network connectivity into 88 components, whereas removing an equal number of random blogs leaves recovery almost unaffected (F1 = 0.874). This asymmetry, together with a counter-intuitive rise in internal modularity under targeted removal, demonstrates that a small set of high-authority "bridge" blogs are structurally responsible for binding the two political communities into a single, politically coherent network.**

***Index Terms*—political blogs network, PageRank, betweenness centrality, Louvain community detection, network robustness, homophily**

I. Introduction and Problem Statement

Social media and blog networks are frequently assumed to reflect real-world social divisions through their link structure. The Political Blogs dataset, compiled by Adamic and Glance [1], provides an unusually strong test of this assumption: 1,490 political blogs were manually labeled as liberal or conservative prior to the 2004 U.S. presidential election, and the hyperlinks between them were recorded independently of these labels. This separation between structure (hyperlinks) and ground truth (political labels) makes the dataset well suited to asking three questions, which form the problem statement of this assignment: (1) how do influence and brokerage appear in the hyperlink structure of political blogs; (2) does the link structure, on its own, recover the two known political orientations; and (3) how do the network's most central blogs affect the stability and quality of the communities that emerge.

This report addresses these questions in six stages, following the assignment workflow: graph construction and cleaning (Section II), centrality selection and analysis (Section III), a centrality-informed removal experiment (Section IV), community detection (Section V), community evaluation against ground truth (Section VI), and a concluding discussion of limitations (Section VII).

II. Dataset and Graph Construction

A. Data Model

The raw dataset (`polblogs.gml`) represents each political blog as a node and each hyperlink from one blog to another as a directed edge. Every node carries a binary attribute `value`, which encodes political orientation as documented by the original compilers and independently confirmed via the University of Florida Sparse Matrix Collection's description of this dataset [4]: a value of 0 indicates a liberal (left-leaning) blog, and a value of 1 indicates a conservative (right-leaning) blog. This label is preserved throughout preprocessing and centrality computation but is never used to construct, tune, seed, or select communities – it is reserved exclusively for the evaluation stage described in Section VI, as required by the assignment.

The graph is modeled as **directed**, because a hyperlink from blog A to blog B represents a specific, one-way act of attention or citation from A toward B, and this asymmetry is meaningful: it is precisely what PageRank exploits to measure authority (Section III). The graph is treated as a **simple graph** rather than a multigraph, because for this analysis only the presence or absence of a link between two blogs is relevant, not how many times one blog links to another; duplicate parallel edges were therefore collapsed rather than retained as edge weights. For the same reason, the graph is treated as **unweighted**: link frequency is not used as a signal of relationship strength anywhere in this analysis.

B. Preprocessing Log

Loading the raw GML file with NetworkX revealed that it is stored as a directed multigraph. The following preprocessing steps were applied in order, with graph size reported before and after each step:

1) **Duplicate edge collapse.** The raw multigraph contained 65 duplicate (parallel) directed edges. These were collapsed into single edges by converting the `MultiDiGraph` to a simple `DiGraph`, retaining one

edge per ordered node pair and discarding no edge weighting, since none was used.

2) **Self-loop removal.** 3 self-loop edges (a blog linking to itself) were identified and removed, as they carry no information about relationships between distinct blogs.
3) **Missing-attribute check.** All 1,490 nodes were confirmed to carry a valid `value` attribute; zero nodes were missing this label.
4) **Isolate removal.** 266 isolated nodes (blogs with zero in-degree and zero out-degree) were removed, since isolated nodes cannot participate meaningfully in centrality or community-detection computations.
5) **Component restriction.** The cleaned graph contained 2 weakly connected components. Analysis was restricted to the **largest weakly connected component**, containing 1,222 of the 1,490 usable nodes (82%), because centrality measures based on shortest paths (e.g. Betweenness, Closeness) are undefined or misleading when computed across disconnected components. This final graph, referred to throughout as the *analysis graph*, is directed and contains 19,021 edges.

The analysis graph additionally contains 422 strongly connected components, indicating that although the great majority of blogs belong to one connected discourse space in the weak (undirected-reachability) sense, most pairs of blogs are not mutually reachable by following hyperlink direction in both directions – reflecting a one-directional attention pattern in which smaller blogs link to major hubs far more often than hubs link back.

C. Graph Properties

Table I summarizes the properties of the final analysis graph.

TABLE I: Analysis Graph Summary Statistics

Metric	Value
Nodes	1,222
Edges	19,021
Density	0.01275
Weakly connected components	1
Strongly connected components	422
Average in-degree	15.57
Average out-degree	15.57

Table II presents the data dictionary for the cleaned analysis graph.

TABLE II: Data Dictionary

Type	Name	Description
Node	id	Integer node identifier
Node	value	Political orientation: 0 = liberal, 1 = conservative
Node	blog_url	Original blog identifier / URL
Edge	(none)	Unweighted directed hyperlink from source to target blog

D. Degree Distribution and Network Structure

Fig. 1 shows the in-degree and out-degree distributions of the analysis graph on a log frequency scale. Both distributions are strongly heavy-tailed: the majority of blogs receive fewer than 10 incoming links, while a small number of hub blogs receive over 100, with the single highest in-degree exceeding 300. This power-law-like shape indicates that influence in the network is highly concentrated in a small number of blogs rather than evenly distributed.

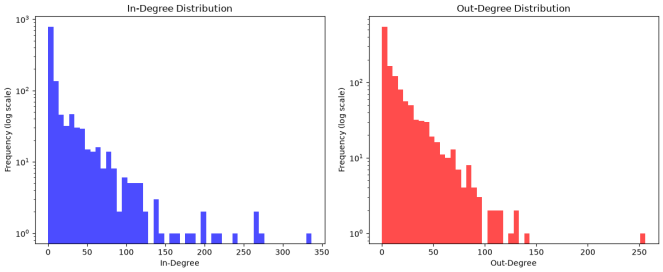


Fig. 1: In-degree and out-degree distributions of the analysis graph (log-scale frequency axis).

Fig. 2 visualizes the top 150 blogs by total degree, colored by political orientation. Even prior to any formal community detection, the visualization shows a clear visual separation between the two political orientations, with only a thin set of edges crossing between the liberal (blue) and conservative (red) clusters. This is direct visual evidence of *homophily* – the tendency of blogs to link preferentially within their own political side – consistent with prior findings on this dataset [1].

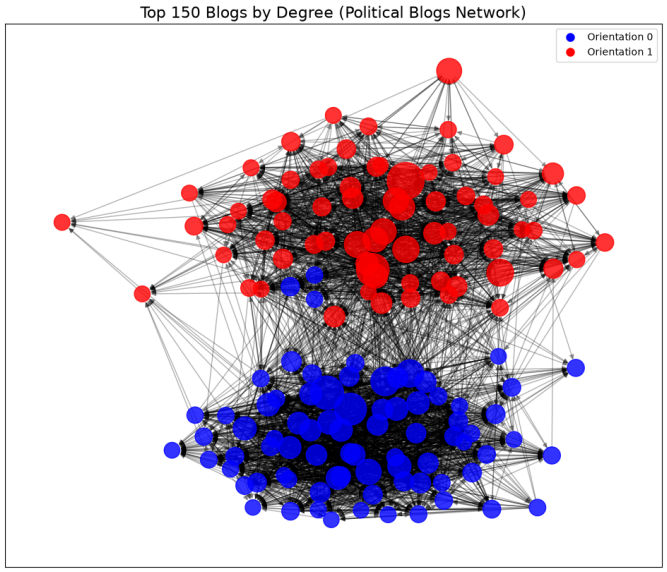


Fig. 2: Top 150 blogs by degree, colored by political orientation (blue = liberal/0, red = conservative/1). Node size is proportional to total degree.

III. Centrality Selection and Analysis

Two centrality measures were selected from the taught list: **PageRank** and **Betweenness Centrality**. Both are valid choices for a directed hyperlink network, but they answer fundamentally different network questions and were chosen specifically for this complementarity.

A. PageRank: Authority

Network question: which blogs are seen as most authoritative by the rest of the network? PageRank is designed for directed link networks in which a link is treated as a directed endorsement, and it identifies blogs that are authoritative because other important blogs link to them. This is preferred over plain in-degree because it accounts for the recursive importance of *who* links to a blog, not merely how many links arrive. PageRank was computed with damping factor $\alpha = 0.85$, the standard value from the original PageRank formulation [2], representing an 85% probability of following an outgoing link versus a 15% random jump. The computation used NetworkX's power-iteration implementation (default `max_iter`=100, `tol`=10^{-6}) and converged successfully without exception.

B. Betweenness Centrality: Brokerage

Network question: which blogs act as structural bridges connecting different parts of the network? Betweenness centrality identifies blogs that lie on many shortest paths between other blog pairs, and can reveal cross-cutting "broker" blogs that plain popularity-based measures (degree, PageRank) cannot detect. Betweenness was computed exactly (not sampled/approximated) with `normalized=True`, so that scores are comparable across graphs of different sizes; the computation considers directed shortest paths, and node pairs with no directed path between them simply contribute zero to betweenness rather than causing an error. Exact computation completed in a few seconds on the 1,222-node analysis graph.

C. Top-10 Rankings

Table III and Table IV report the top-10 ranked blogs for each measure.

TABLE III: Top 10 Blogs by PageRank

Rank	Blog	PageRank Score
1	dailykos.com	0.0189
2	atrios.blogspot.com	0.0160
3	instapundit.com	0.0133
4	blogsforbush.com	0.0132
5	talkingpointsmemo.com	0.0131
6	michellemalkin.com	0.0115
7	drudgereport.com	0.0113
8	washingtonmonthly.com	0.0111
9	powerlineblog.com	0.0094
10	andrewsullivan.com	0.0091

TABLE IV: Top 10 Blogs by Betweenness Centrality

Rank	Blog	Betweenness Score
1	blogsforbush.com	0.1467
2	atrios.blogspot.com	0.0611
3	instapundit.com	0.0512
4	dailykos.com	0.0369
5	newleftblogs.blogspot.com	0.0308
6	madkane.com/notable.html	0.0302
7	wizbangblog.com	0.0273
8	lashawnbarber.com	0.0243
9	hughhewitt.com	0.0230
10	washingtonmonthly.com	0.0219

D. Comparison of Rankings

A Spearman rank correlation between the full PageRank and Betweenness rankings (all 1,222 nodes) yields $\rho = 0.758$ ($p \ll 0.001$), indicating a strong but imperfect relationship between the two measures. Exactly 5 of the top-10 blogs by each measure overlap: `dailykos.com`, `atrios.blogspot.com`, `instapundit.com`, `washingtonmonthly.com`, and `blogsforbush.com`. This shows that some blogs are simultaneously authoritative and structurally central, while others specialize in one role.

E. Node-Level Interpretation

dailykos.com is the top-ranked blog by PageRank, consistent with its well-documented real-world role as one of the most prominent liberal political blogs of the 2004 era, widely cited as an authoritative source by other blogs. **blogsforbush.com** is the top-ranked blog by Betweenness despite ranking only 4th by PageRank – this shows it functions primarily as a structural bridge sitting on many shortest paths between other blogs, rather than as the single most broadly cited authority. **michellemalkin.com** is highly ranked by PageRank (6th) but does not appear in the Betweenness top-10 at all, illustrating that high authority does not guarantee a brokerage role: some highly cited blogs sit within one ideological cluster rather than bridging between clusters. Together, these three cases demonstrate that a high centrality score alone does not establish a uniform notion of "importance" – the specific measure used determines what kind of importance is being captured.

IV. Centrality-Informed Community Experiment

To connect the centrality analysis to the community-detection stage, a controlled node-removal experiment was designed using **PageRank** as the intervention measure. PageRank was chosen over Betweenness for this role because it represents overall authority and influence in the network, directly connecting to the assignment's problem statement focus on "influence."

A. Experimental Design

The top 5% of nodes by PageRank score ($n = 61$ of 1,222 nodes) were removed to create a *targeted-removal graph*. This threshold was chosen to be large enough to produce a

measurable structural effect while retaining the vast majority of the network for meaningful analysis. As a control, 61 nodes were also removed uniformly at random (fixed seed = 42, using Python's `random.sample`) to create a *random-removal graph* of identical size. Both experimental graphs were derived from the same analysis graph using `.copy()`, ensuring the original graph remained unmodified for baseline comparison.

B. Structural Comparison

Table V compares the three graphs' structural properties.

TABLE V: Structural Comparison: Original vs. Targeted vs. Random Removal

Metric	Original	Targeted	Random
Nodes	1,222	1,161	1,161
Edges	19,021	10,279	17,404
Density	0.01275	0.00763	0.01292
Weakly conn. comp.	1	88	4
Largest comp. size	1,222	1,071	1,155

Removing only 5% of nodes by PageRank eliminated 46% of all edges (19,021 $\rightarrow$ 10,279), while removing the same number of random nodes eliminated only 8.5% of edges (19,021 $\rightarrow$ 17,404) – a direct consequence of hub nodes carrying disproportionately many edges. The targeted-removal graph fragmented into 88 weakly connected components (largest component reduced by 12.4%), whereas the random-removal graph fragmented into only 4 components (largest component reduced by only 5.5%). This confirms that the network is substantially more fragile to the loss of high-authority hub blogs than to the loss of random blogs, mirroring a well-established finding in network science that scale-free networks tend to be robust against random failure but fragile against targeted attacks on their hubs [3].

V. Community Detection

Louvain modularity maximization was selected as the community-detection method. This method assumes **disjoint, non-overlapping** communities of variable size, and does not require the number of communities to be specified in advance, which is well suited to this network where the number of natural clusters is not known a priori. Louvain does not natively operate on directed graphs; each directed graph was therefore converted to an undirected graph by taking the **union** of edges in both directions – an undirected edge exists between two blogs if a hyperlink exists between them in either direction. This transformation was chosen because a hyperlink in either direction is treated as evidence of a topical relationship between two blogs, regardless of its originating direction. Because Louvain's outcome depends stochastically on internal iteration order, it was run with 5 fixed seeds (42–46) on each of the three graphs (original, targeted, random), and results are reported as mean $\pm$ standard deviation across seeds; the seed-42 partition was declared for all visualizations and profiling.

A. Community Counts and Modularity

Table VI summarizes the community-detection results across all three graphs.

TABLE VI: Community Detection: Three-Way Comparison (5 seeds)

Metric	Original	Targeted	Random
Mean # communities	10.8	101.2	14.0
Std # communities	0.40	1.47	0.00
Mean modularity	0.4269	0.4583	0.4239
Std modularity	0.0001	0.0015	0.0001
Largest comm. size	635	468	602
Singleton comms.	0	84	2

The extremely low standard deviation in modularity across seeds (all $\leq$ 0.0015) indicates that Louvain converges to essentially the same partition quality regardless of seed for this network, i.e. the community structure is stable and not an artifact of a favorable random initialization.

B. Full Community Size Distribution

Beyond the aggregate counts in Table VI, the full size distribution of the declared (seed-42) partition for each graph was examined: Original – 11 communities with sizes [635, 526, 40, 4, 4, 3, 2, 2, 2, 2, 2]; Targeted – 100 communities with sizes [468, 399, 169, 13, 4, 3, 3, 2$\times$8, then 90 singleton communities of size 1]; Random – 14 communities with sizes [602, 493, 39, 4, 4, 3, 3, 3, 2, 2, 2, 2, 1, 1]. Fig. 3 visualizes these distributions. The Original and Random graphs both show a clean "two giants plus a short tail" shape, characteristic of a core-periphery structure with two dominant political blocs. The Targeted graph shows the same two-to-three dominant communities, but is followed by a long, dense staircase of roughly 90 singleton or near-singleton fragments – visual confirmation that removing high-authority hubs does not merely increase the community *count*, but shatters the network's periphery into isolated pieces.

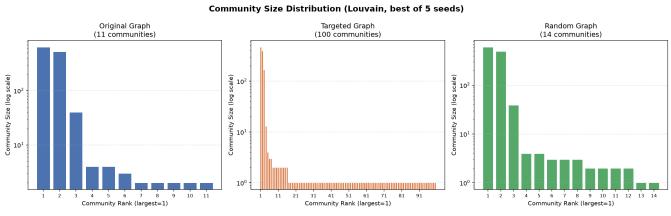


Fig. 3: Full community size distribution (log scale) for the original, targeted-removal, and random-removal graphs, ranked from largest to smallest.

C. Community Visualization and Profiling

Fig. 4 shows the top-150-by-degree subgraph colored by detected community (original graph, seed 42), showing a striking correspondence to the political-orientation visualization of Fig. 2. Table VII profiles the five largest detected communities using the two centrality measures from Section III.

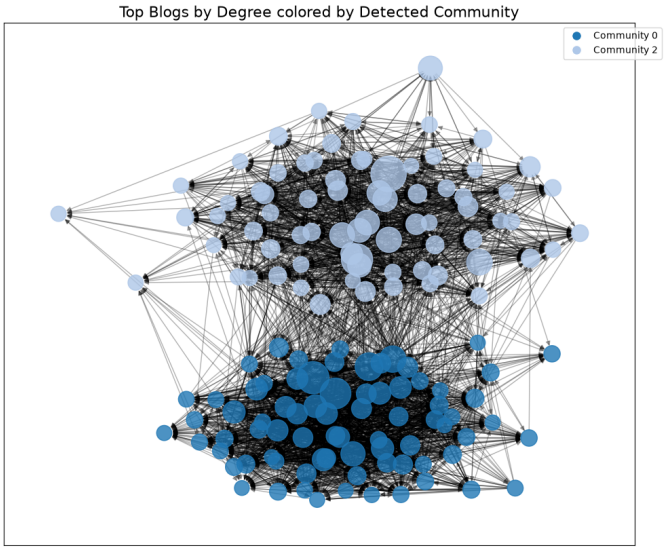


Fig. 4: Top blogs by degree, colored by Louvain-detected community (original graph, seed 42).

TABLE VII: Top 5 Community Profiles (Original Graph, Seed 42)

Comm.	Size	Mean PR	Mean Betw.	Purity
2	635	0.000825	0.001248	94.8% (Cons.)
0	526	0.000869	0.001459	97.5% (Lib.)
4	40	0.000318	0.000239	70.0% (Lib.)
5	4	0.000283	0.000239	75.0% (Lib.)
9	4	0.000569	0.000688	100.0% (Cons.)

Louvain detected an average of 10.8 communities on the original graph – more than the two known political orientations – indicating that link structure divides the network at a finer granularity than the political binary alone, likely reflecting topic-level sub-clusters within each political side. The two largest communities are already highly politically homogeneous (94.8% and 97.5% purity) before any formal evaluation against ground truth, strongly suggesting that link structure closely tracks political alignment.

VI. Community Evaluation

Detected community IDs are arbitrary integers with no inherent correspondence to the 0/1 ground-truth labels (the *label-permutation problem*); comparing community ID 0 to label 0 directly would be meaningless, since Louvain could equally have assigned that community the ID 5. To avoid this, communities are evaluated using a **pairwise** methodology: for every unordered pair of nodes (i, j) in a graph, we check whether i and j are in the same detected community, and separately whether they share the same ground-truth orientation. This sidesteps the need to know which arbitrary community ID corresponds to which political side, since only relative agreement between pairs is measured. Table VIII restates the exact pairwise definitions used, matching the assignment specification.

TABLE VIII: Pairwise Evaluation Definitions

Term	Pairwise meaning
TP	Same detected community, same ground-truth orientation
TN	Different detected communities, different orientation
FP	Same detected community, different orientation
FN	Different detected communities, same orientation

A. Results

Table IX through Table XI report the pairwise confusion matrices for the three graphs, and Table XII summarizes all four external metrics.

TABLE IX: Confusion Matrix – Original Graph (746,031 pairs)

	Same Orient.	Diff. Orient.
Same Community	TP = 313,296	FP = 26,874
Diff. Community	FN = 60,039	TN = 345,822

TABLE X: Confusion Matrix – Targeted Graph (673,380 pairs)

	Same Orient.	Diff. Orient.
Same Community	TP = 184,576	FP = 18,398
Diff. Community	FN = 152,376	TN = 318,030

TABLE XI: Confusion Matrix – Random Graph (673,380 pairs)

	Same Orient.	Diff. Orient.
Same Community	TP = 279,840	FP = 23,105
Diff. Community	FN = 57,430	TN = 313,005

No denominator was zero for any of the three graphs' metrics; had this occurred (e.g. a graph with no same-community pairs), the corresponding metric would have been reported as *undefined* rather than silently replaced with zero, per the zero-denominator rule.

B. Internal vs. External Quality

Standard modularity is a valid internal-quality measure here because Louvain produces disjoint, non-overlapping communities, satisfying the assumption required by the standard modularity formula. Table XIII juxtaposes internal modularity against the external accuracy/F1 metrics.

On the original graph, link structure alone recovers political orientation with high fidelity (accuracy = 0.8835, F1 = 0.8782), achieved entirely without the detection process ever observing the political labels. Comparing across graphs reveals the central finding of this study: the targeted-removal graph achieves the *highest* modularity of all three graphs (0.4583)

TABLE XII: External Evaluation Summary

Graph	Acc.	Prec.	Rec.	F1
Original	0.8835	0.9210	0.8392	0.8782
Targeted	0.7464	0.9094	0.5478	0.6837
Random	0.8804	0.9237	0.8297	0.8742

TABLE XIII: Internal Modularity vs. External Accuracy/F1

Graph	Modularity	Accuracy	F1
Original	0.4269	0.8835	0.8782
Targeted	**0.4583**	0.7464	**0.6837**
Random	0.4239	0.8804	0.8742

yet the *lowest* F1 score (0.6837) – a striking dissociation between internal structural quality and external political-recovery accuracy. Removing the cross-partisan "bridge" blogs allows Louvain to carve out tighter, more internally dense clusters that no longer track the real political boundaries as well. Random removal, by contrast, barely affects political recovery (F1 = 0.8742, a drop of only 0.004 from baseline), while targeted removal causes a 0.194 drop in F1. This directly echoes the fragility pattern identified structurally in Section IV and via community proliferation in Section V: the network's ability to encode recoverable political structure depends heavily on a small set of high-authority bridge blogs, not on the network as a whole. Strong structural communities (high modularity) can therefore still disagree with political orientation, because internal density and external label alignment are fundamentally different properties that can move in opposite directions when the underlying bridging structure is disrupted.

VII. Conclusion, Limitations, and Reflection

A. Conclusion

The political blog hyperlink network from the 2004 U.S. election cycle exhibits strong latent political structure recoverable through link topology alone. Louvain community detection on the original graph identified approximately 10–11 communities per run, converging on two dominant groups of sizes 635 and 526 with purities of 94.8% and 97.5% respectively – nearly perfectly aligned with the known liberal/conservative ground truth. The pairwise evaluation confirms this quantitatively (accuracy = 0.8835, F1 = 0.8782). Crucially, the centrality-informed removal experiment reveals that this recoverability is *fragile* to the loss of the network's most authoritative nodes: removing just 5% of blogs selected by PageRank reduced F1 from 0.8782 to 0.6837, fragmented connectivity from 1 into 88 weakly connected components, and caused Louvain to detect roughly 101 communities on average (versus $\sim$11 for the original graph) – while an equivalently-sized random removal left F1 nearly unchanged (0.8742) and connectivity/community counts close to the original (4 components, $\sim$14 communities). This asymmetry, together with the counter-intuitive rise in modularity under targeted removal, shows that a small number of high-authority hub blogs function not only as authorities in the PageRank sense, but as the structural bridges that bind the two political communities into a single, coherent, and politically recoverable network.

B. Limitations

- **Temporal snapshot:** This dataset is a snapshot of approximately 1,500 political blogs from the 2004 U.S. presidential election cycle. The hyperlink patterns, blog prominence, and discourse structures of that era may not generalize to modern social media platforms, later elections, or non-U.S. political contexts.
- **Hyperlinks are not endorsements:** A directed hyperlink from blog A to blog B does not necessarily indicate agreement. Blogs frequently link to ideological opponents to criticize or rebut them. The link structure therefore conflates agreement-based and adversarial linking, which may partially explain the false positives and false negatives observed in the pairwise evaluation.
- **Louvain stochasticity:** Although a fixed random seed was used for the declared partition, Louvain's stochastic phase-ordering means results can vary slightly between runs and library versions, even with identical seeds. The 5-seed stability check showed very low modularity variance (std $\leq$ 0.0015), but node-level community membership may not be fully reproducible across library upgrades.
- **Centrality-measure dependence:** The definition of "importance" used to drive the removal experiment depends entirely on the chosen centrality measure. PageRank was used as the intervention measure; using Betweenness, Eigenvector, or in-degree centrality instead would select a different removal set and could yield different fragility conclusions. The top-10 overlap between PageRank and Betweenness was only 5 of 10 nodes (Spearman $\rho = 0.758$), confirming these measures capture genuinely different aspects of node importance.

References

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